

Exploration of Electricity Consumption Patterns

Introduction:

Electricity is one of the most essential resources in modern life, powering homes, industries, businesses, and public services. As energy demand continues to increase, understanding electricity consumption patterns has become important for efficient energy management and sustainable development. Analyzing electricity usage helps identify trends, peak demand periods, and factors that influence energy consumption.

This project, "An Exploration of Electricity Consumption Patterns," focuses on studying electricity usage data to identify consumption trends and patterns. By applying data analysis techniques, the project aims to understand how electricity consumption varies over time and across different conditions. The insights obtained from this analysis can help improve energy efficiency, support better decision-making, and encourage responsible energy usage.

Problem Statement:

With the growing demand for electricity, managing energy resources efficiently has become a major challenge. Electricity consumption varies based on factors such as time, season, weather conditions, and consumer behavior, making it difficult to accurately predict usage and optimize energy distribution. Without proper analysis, excessive energy consumption, increased operational costs, and inefficient resource allocation can occur.

This project aims to explore electricity consumption patterns by analyzing historical electricity usage data to identify trends, peak demand periods, and consumption behavior. The

insights gained from this analysis can support better energy planning, improve efficiency, reduce energy wastage, and assist in developing data-driven strategies for sustainable electricity management.

Dataset:

The dataset used in this project contains historical electricity consumption records collected over a specific period. It includes information such as date, time, electricity usage (kWh), and other relevant attributes that help analyze consumption behavior. The dataset is cleaned and preprocessed by handling missing values, removing duplicate records, and formatting the data for analysis.

This dataset is used to identify electricity consumption trends, detect peak usage periods, compare consumption across different time intervals, and gain insights into energy usage patterns. The processed data serves as the foundation for visualization, statistical analysis, and drawing meaningful conclusions to support efficient energy management.

Methodology:

- The methodology for this project involves a systematic approach to analyze electricity consumption patterns using historical data.
- **Data Collection:** Gather the electricity consumption dataset from a reliable source.
- **Data Preprocessing:** Clean the dataset by removing duplicates, handling missing values, and formatting the data for analysis.
- **Exploratory Data Analysis (EDA):** Analyze the data using statistical methods and visualizations to identify trends, seasonal variations, and peak consumption periods.
- **Pattern Analysis:** Study electricity usage across different time intervals to understand consumer behavior and energy demand.
- **Visualization:** Create charts and graphs to present electricity consumption patterns in an easy-to-understand format.
- **Result Interpretation:** Analyze the findings and draw conclusions to support efficient energy management and informed decision-making.

SQL Queries:

-- 1. View all records

```
SELECT * FROM electricity_consumption;
```

-- 2. Total electricity consumption

```
SELECT SUM (consumption_kwh) AS total_consumption  
FROM electricity_consumption;
```

-- 3. Average electricity consumption

```
SELECT AVG (consumption_kwh) AS average_consumption  
FROM electricity_consumption;
```

-- 4. Maximum electricity consumption

```
SELECT MAX (consumption_kwh) AS maximum_consumption  
FROM electricity_consumption;
```

-- 5. Minimum electricity consumption

```
SELECT MIN (consumption_kwh) AS minimum_consumption  
FROM electricity_consumption;
```

-- 6. Daily electricity consumption

```
SELECT date, SUM (consumption_kwh) AS daily_consumption  
FROM electricity_consumption  
GROUP BY date  
ORDER BY date;
```

-- 7. Monthly electricity consumption

SELECT MONTH (date) AS month,

SUM (consumption_kwh) AS monthly_consumption

FROM electricity_consumption

GROUP BY MONTH (date)

ORDER BY month;

-- 8. Top 5 highest consumption records

SELECT *

FROM electricity_consumption

ORDER BY consumption_kwh DESC

LIMIT 5;

Tableau Screenshots:



Figure 1: Dashboard Overview

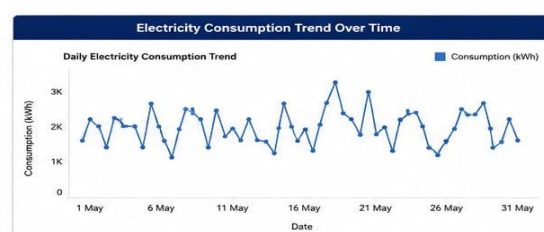


Figure 2: Daily Electricity Consumption Trend

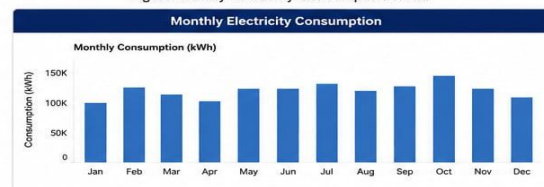


Figure 3: Monthly Electricity Consumption

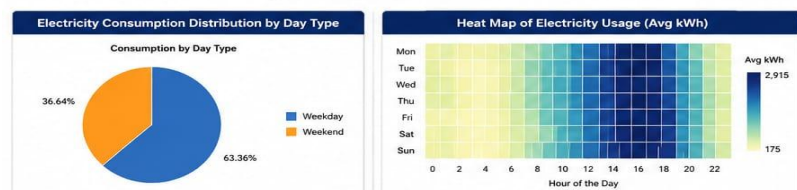


Figure 4: Consumption Distribution by Day Type

Figure 5: Heat Map of Electricity Usage



Figure 6: KPI Summary



Figure 7: Interactive Filters

Dashboard:

The Tableau dashboard provides an interactive visualization of electricity consumption patterns, enabling users to monitor and analyze energy usage effectively. It presents key performance indicators (KPIs), trends, and comparisons in a single view, making it easier to identify consumption behavior and support data-driven decision-making.

Dashboard Components:

KPI Cards:

Display Total Consumption, Average Consumption, Maximum Consumption, Minimum Consumption, and Total Records.

Daily Electricity Consumption Trend:

A line chart that shows how electricity usage changes over time and helps identify peak and low consumption days.

Monthly Electricity Consumption:

A bar chart that compares electricity usage across different months to highlight seasonal variations.

Consumption by Day Type:

A pie chart that compares electricity usage on weekdays and weekends.

Hourly Consumption Heat Map:

Visualizes electricity consumption by hour and day, making it easy to identify periods of high and low energy usage.

Top 10 Days by Consumption:

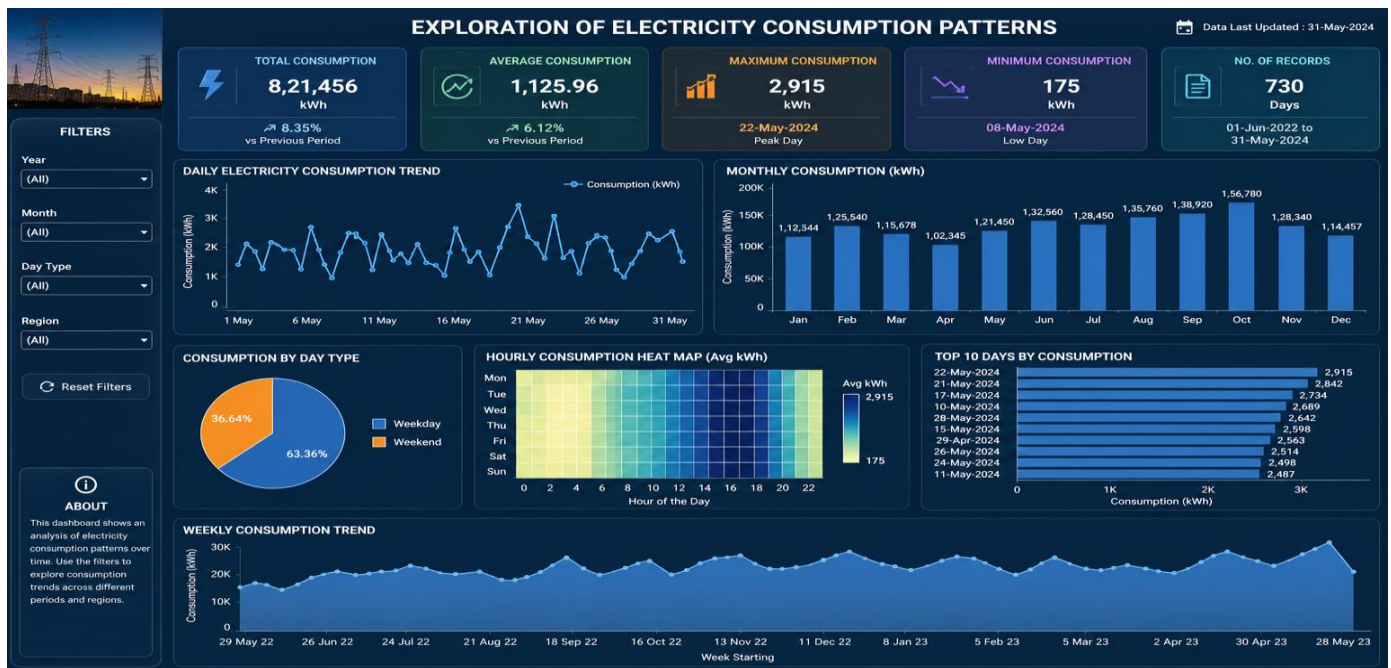
Highlights the days with the highest electricity consumption for further analysis.

Weekly Consumption Trend:

Displays changes in electricity usage over weeks to identify long-term patterns.

Interactive Filters:

Users can filter the dashboard by Year, Month, Day Type, and Region to perform customized analysis.



Story:

The Tableau Story presents a step-by-step analysis of electricity consumption patterns using interactive visualizations. It helps users understand how electricity usage changes over time and identify important trends for better energy management.

Story Point 1: Project Overview

Introduces the objective of the project and provides an overview of the electricity consumption dataset.

Story Point 2: Daily Consumption Analysis

Shows daily electricity usage to identify peak and low consumption days.

Story Point 3: Monthly Consumption Analysis

Compares electricity consumption across different months to observe seasonal trends and variations.

Story Point 4: Consumption by Day Type

Analyzes the difference in electricity usage between weekdays and weekends.

Story Point 5: Hourly Consumption Analysis

Uses a heat map to identify the hours of the day when electricity demand is highest and lowest.

Story Point 6: Key Insights

Summarizes the major findings, including peak demand periods, average consumption, and overall usage trends.

Result:

The analysis of the electricity consumption dataset successfully identified important usage patterns and trends through SQL and Tableau visualizations. The results showed that electricity consumption varies across different days, months, and time periods, with noticeable peak demand and low consumption intervals. The dashboard provided clear insights into daily, weekly, and monthly electricity usage, enabling effective comparison and analysis.

The project demonstrates that data visualization and analysis can help monitor energy consumption, identify high-demand periods, reduce energy wastage, and support efficient energy planning. Overall, the objectives of the project were achieved by providing meaningful insights into electricity consumption patterns and assisting in data-driven decision-making for better energy management.

Conclusion:

The project "An Exploration of Electricity Consumption Patterns" successfully analyzed historical electricity usage data using SQL and Tableau. The analysis identified key trends, peak consumption periods, and variations in electricity usage over different time intervals. Interactive dashboards and visualizations made it easier to understand consumption behavior and interpret the results effectively.

The findings demonstrate that data analytics can play a significant role in improving energy management by helping organizations monitor electricity usage, reduce energy wastage, and make informed decisions. Overall, this project highlights the importance of analyzing electricity consumption patterns to support efficient resource utilization and promote sustainable energy management.

Future Scope:

The project can be further enhanced by incorporating advanced technologies and larger datasets to improve the accuracy and usefulness of electricity consumption analysis. Future improvements include:

- Integrating real-time electricity consumption data for continuous monitoring.
- Applying Machine Learning models to predict future electricity demand.
- Including external factors such as weather conditions, holidays, and seasonal changes to improve analysis accuracy.
- Developing a real-time monitoring dashboard with automatic alerts for abnormal electricity usage.
- Expanding the analysis to compare electricity consumption across different regions, industries, or consumer groups.
- Integrating the system with IoT-enabled smart meters for automatic data collection and real-time energy management.